

EDUCATIONPLUS

GET THE EDGE Follow us facebook.com/thehindu x.com/the_hindu instagram.com/the_hindu

John J. Kennedy

India's higher education story over the past decade has been one of transformation. Women are entering universities in unprecedented numbers. In several disciplines, enrolment has crossed 50%. The gender gap, by many measures, is closing. Engineering is the exception.

Let's look at numbers. Women account for roughly 28-30% of undergraduate Engineering enrolment, according to the All India Survey on Higher Education 2021-22. That number has barely moved in years. While overall engineering enrolment expanded significantly in the early 2010s, women's share didn't keep pace. The rest of higher education tilted toward gender parity. Engineering stayed still.

But even that number flattens the reality. Women in Engineering are heavily concentrated in Computer Science and Electronics. In Mechanical, Civil, and Electrical Engineering, which are the backbone disciplines, female enrolment is often in the single digits at many institutions. So it's not just that women are under-represented in Engineering broadly. Within Engineering itself, they are being funnelled into a narrow slice of the field.

Beyond graduation, the numbers fall off a cliff. Women make up close to a third of Engineering graduates but less than one-fifth of the research and development workforce in Science and Engineering. Somewhere between the degree and the job, a large proportion of women exit and



GETTY IMAGES/ISTOCKPHOTO

Out of circuit

Why women aren't staying in Engineering and what needs to be done

don't come back. This is the "leaky pipeline" problem and it shows up at nearly every stage.

The easy explanation is individual preference. Women choose other fields or they opt out. But that explanation doesn't hold up well under

scrutiny, because choices don't happen in a vacuum.

A closer look

In rural and semi-urban settings, girls consistently receive less encouragement in Maths and Science. Access to coaching, which is practically a pre-

requisite for India's competitive Engineering entrance exams, skews heavily toward boys, constrained by both money and mobility. Even at the IITs, where active interventions have increased female enrolment in recent years, the proportion has stabilised at

around one-fifth. The floor has risen, but the ceiling is staying put.

Then there's the economics of household decision-making. Engineering is expensive. Families making hard choices about where to invest often do so along deeply gendered lines; sons toward high-cost, high-return professional degrees; daughters toward disciplines perceived as safer, more flexible, more compatible with the life she's expected to lead later. These decisions aren't made maliciously. They're made pragmatically, within a set of assumptions that nobody stops to question.

Interestingly, the workplace is where the pipeline leaks most visibly. Core Engineering sectors such as manufacturing, construction, and heavy industry remain stubbornly male-dominated, and not just in perception. Safety infrastructure is often inadequate. Work-life flexibility is minimal. Career advancement is slow and opaque. Women who graduate with Engineering degrees and enter these environments often migrate toward non-technical roles or leave the field altogether.

There's also the absence of role models, which matters more than people usually credit. Women are under-represented in Engineering faculty, in industry leadership, and in the rooms where technical decisions get made. That under-representation isn't just a statistic; it shapes what younger women imagine is possible for themselves. When you rarely see someone who looks like you succeeding in a space, it's rational to wonder whether that space is really for you.

What makes this paradoxical

is that India actually performs reasonably well in terms of women's overall participation in STEM. The aggregate numbers look decent by international standards. But those numbers obscure a stark internal divide: women entering Science in large numbers, while Engineering, especially its core branches, remains a different story entirely. Policy so far has focused mainly on access: scholarships, outreach, and reserved seats. These matter, but they're not enough. Getting more women into Engineering programmes doesn't solve the problem if the structures governing what happens next remain unchanged.

Looking ahead

So, what's actually needed is intervention across the full arc: school through career. Gender-sensitive teaching in Maths and Science. Affordable preparatory pathways that don't require a family to pay for private coaching. University environments that don't subtly channel women into a handful of "acceptable" specialisations.

Most critically, workplaces that are actually built for a diverse workforce: flexible structures, real anti-harassment mechanisms, clear re-entry pathways for women returning after breaks. The deeper issue isn't access. It's that the entire system, from classroom to boardroom, is still configured around assumptions that don't include women as full participants in engineering. Until that changes, the problem won't.

The writer is former professor and dean, Christ University, Bengaluru

SCHOLARSHIPS

University of Dundee GEMS Scholarship
Offered by the University of Dundee, the U.K.
Eligibility: Students who have secured an offer for a full-time UG course with a tuition fee of at least £21,600 per annum at the GEMS Schools for September 2026 and have qualified for international fee paying status
Rewards: £6,000 annually
Application: Online
Deadline: August 28
www.b4s.in/edge/UGEM2

NSP Pre-Matric Scholarship for Students with Disabilities
Offered by the Ministry of Social Justice and Empowerment, Government of India.
Eligibility: Indian nationals pursuing regular full-time studies in Class 9 or 10 in a recognised school with a disability of 40% or more as defined under the Rights of Persons with Disabilities Act, 2016, and an annual family income of less than ₹250,000.
Rewards: Up to ₹4,000 and benefits.
Application: Online
Deadline: August 31
www.b4s.in/edge/PMSTD7

Nokia Foundation Scholarship
Offered by the Nokia Foundation
Eligibility: Students pursuing a doctoral degree in ICT or related scientific fields in a Finnish University with at least one accepted publication related to doctoral dissertation at the time of applying, and a good academic record.
Rewards: €7,500
Application: Online
Deadline: September 15
www.b4s.in/edge/NKSP2

Courtesy: Buddy4study.com



OFF THE EDGE
Nandini Raman

I have completed a PG in Hindi and a PG Diploma in Translation (English and Hindi). I cleared the first level of translation exams conducted by SSC, but not the second. How can I look for any translation or content writing jobs? I hear that people refuse to pay after the work is completed. Parthasaradhi

Dear Parthasaradhi,

Create a LinkedIn profile in both English and Hindi. Register on Naukri.com using keywords such as "Hindi Linguist". Upload your translation portfolio (News, Legal, Creative) to Google Drive and join translation groups on Facebook and Telegram to network. Focus on professional opportunities like the National Translation Mission (NTM), which outsources academic and technical translations.

Prasar Bharati and All India Radio have openings for Casual Translator or News Editor (Hindi). Banks like SBI, NABARD, and RBI recruit Official Language Officers through their own exams.

In the private sector, companies like Lionbridge, RWS, and TransPerfect hire Hindi translators for localisation, website translation, and subtitling. Amazon, Flipkart, and many ed-tech companies recruit translators for Hindi content. With a PG in Hindi, you can also learn basic SEO and explore Hindi content writing or advertising copywriting.

For payments, use trusted platforms like Upwork or Fiverr, where payment is secured before work begins. For independent clients, always ask for a 20-50% advance. Never provide free samples longer than 200 words. Also check ProZ.com

Steer your career

Uncertain about your career options? Low on self-confidence? This column may help

to verify agencies through its Blue Board payment ratings.

The second stage of SSC focuses on translation and essays, so strengthen your contextual translation and grammar. Translate editorials from English into Hindi daily and read the bilingual Official Gazette of India to master the Administrative Hindi used in SSC exams.

I am a final-year B.A. Mass Communication student with English as an allied subject. Over the three years, my total credits in English are only seven. How can I become eligible for further admissions in this subject? Madiha

Dear Madiha,

Most Central Universities require 24-32 credits in English for admission to an M.A. programme. One practical solution is the Additional Subject route. Many universities allow students to appear for additional English Literature papers and issue a separate mark sheet. Also explore B.A. Additional or Single Subject Certification programmes offered through distance education by state universities.

Since 2021, the UGC has allowed students to earn up to 40% of their credits through SWAYAM/NPTEL. You can take English Literature courses, but your college must agree to transfer these credits to your final transcript. As you're in your final year, speak to your Dean or Controller of Examinations immediately. Another option is a P.G. Diploma in English (PGDE), which many universities offer as a bridge to an M.A. You could also pursue a Certificate in Teaching of English (CTE).

Finally, read the CUET PG prospectus carefully. Some universities allow graduates from any discipline to apply for an M.A. in English based on performance in the entrance exam. Finally, consider M.A. Media and Cultural Studies or Film Studies, which complement your Mass Communication background while offering substantial literary and cultural analysis.

My daughter will finish B.Tech ECE in July 2029. What are the other industry-relevant skills she should develop? Madhusudhan

Dear Madhusudhan,

The ECE job market in 2029 will reward T-shaped professionals: those with deep expertise in one core area and a broad understanding of AI and software. The BS in Electronic Systems from IIT-Madras is an excellent addition, provided she can manage the workload. Since she is already pursuing a B.Tech., she can complete it at her own pace. Even exiting with a Diploma will strengthen her resume.

AI/ML should complement — not replace — her core ECE subjects. The future lies in Edge AI, where AI runs directly on devices rather than the cloud. She should learn Python and deploy machine-learning models on microcontrollers like Arduino or ESP32 and focus on practical applications such as building AI-powered smart sensors instead of basic chatbots. To stand out, she should build expertise in three areas: Core Hardware, System Software, and Emerging Technologies.

Since my marks in NEET

were not good, I had to join a private veterinary college. What are my career options after B.V.Sc and AH? Or should I go for higher studies? Gokul

Dear Gokul,

B.V.Sc. and AH is one of India's few professional degrees with excellent employability and little competition from other medical streams. The private college tag matters far less than VCI recognition. During your final-year internship, work at a busy Government Veterinary Hospital to gain hands-on surgical and diagnostic experience. If you want job security, begin preparing for your state's Veterinary Assistant Surgeon (VAS) exam in the third or fourth year.

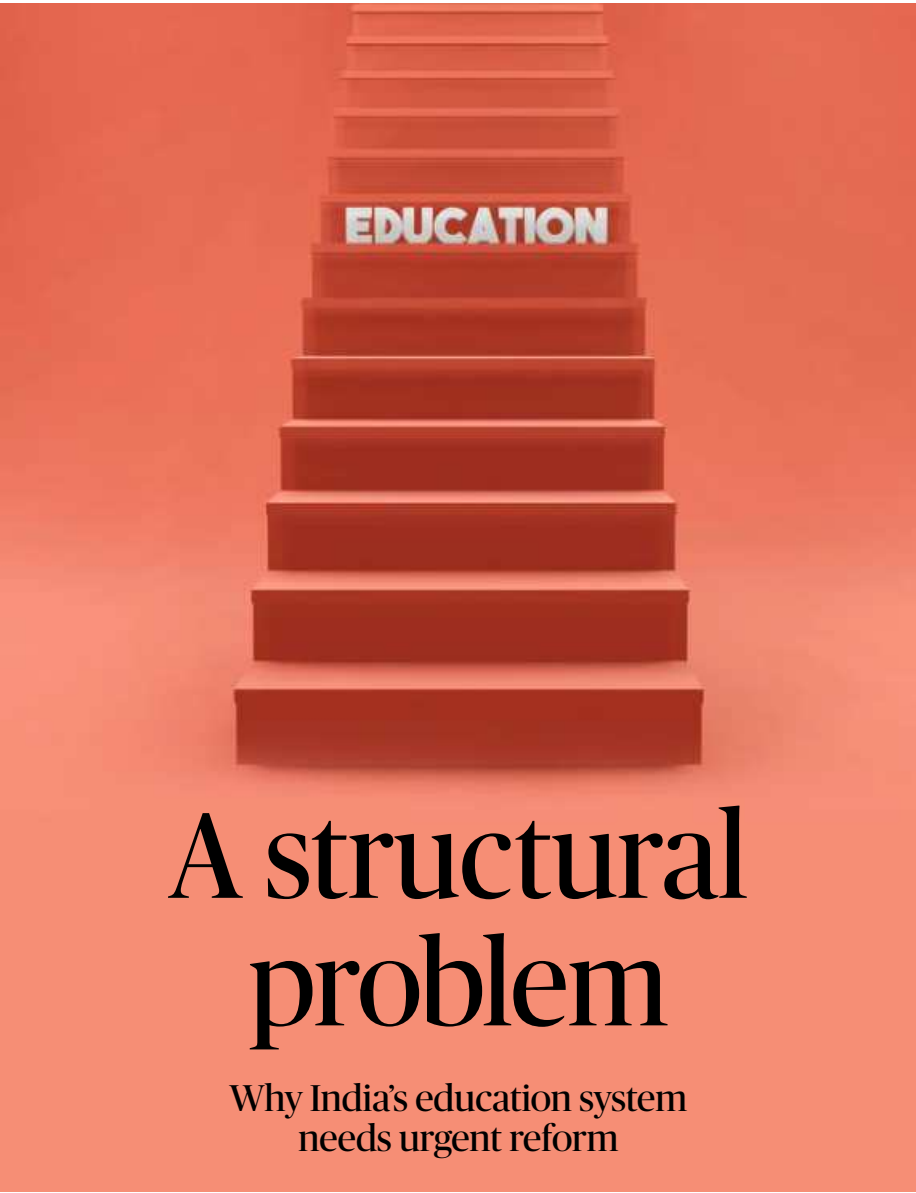
Career options include Veterinary Officer, Livestock Development Officer, Remount Veterinary Corps (Army), Technical Officer in private companies, dairy organisations, Agricultural Field Officer in banks, or starting your own practice.

An M.V.Sc. is worthwhile if you plan to specialise, teach, or pursue research. High-demand specialisations include Surgery, Medicine, Gynaecology, Animal Nutrition, and Pathology. If you enjoy the business side more, consider an MBA in Agri-Business that can lead to managerial roles in the food and animal-health sectors. An MPH can also open opportunities with organisations such as the WHO and FAO.

Veterinarians are also in demand in the U.K., Canada, Australia, and New Zealand, though you'll need to clear their licensing exams before practising.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge



GETTY IMAGES/ISTOCKPHOTO

A structural problem

Why India's education system needs urgent reform

Anil Sarwal

The recent exam paper leaks have rightly provoked public outrage. But treating them as an isolated scandal is to mistake a symptom for the disease. The rot runs far deeper and, until we are willing to confront its full extent, no amount of crisis management will produce lasting change.

Black teaching

India's education system is in structural distress. In 1966, the Kothari Commission recommended that public expenditure on education should reach 6% of GDP. Six decades later, we still hover between 2.9-3.5%; a figure that speaks

not only of neglect but also of a settled national indifference to learning as a public good. The consequences are everywhere visible: crumbling infrastructure, overcrowded classrooms, and schools so starved of resources that even blackboards and functional toilets cannot be taken for granted. The entire educational process in the country can be termed "black teaching", a situation in which learners are deprived of their legitimate share of learning.

Consider a single staggering statistic: India has more than one lakh single-teacher schools, serving more than 33 lakh students. In these schools, one teacher often teaches up to 18 subjects across five

grade levels simultaneously. The Right to Education Act mandates a very different reality. The distance between that mandate and the ground truth is the measure of our collective failure.

The problem does not end with infrastructure. Much classroom instruction remains trapped in the lecture method; passive, rote-driven, and largely indifferent to whether students are actually learning. The evidence is damning. Many students completing Class 8 cannot read a simple text or write a few coherent sentences in their mother tongue. Yet they are passed along, year after year, through a system that has quietly abandoned them.

The writer is a retired Associate Professor of English.

Tavishi Tewary

For decades, the global response to the climate crisis has been dominated by a “hardware-first” mentality. The missing component is climate education as a fundamental climate mitigation measure. Education systems are uniquely positioned to radically redefine long-term emissions trajectories by re-tuning human behaviour, changing consumption patterns, and creating a form of civic engagement with the potential to wield power to hold.

Mismatch

In India, there is still a fundamental mismatch between education system and green jobs aspirations. The institutions charged with the responsibility of creating a green workforce are languishing in abstract, rote-based environmental education, which is mostly out of touch with the actual issues. The majority of individuals concur that climate change is a menace, but very few have the sys-

Agents of change

Why India's sustainability education must move from theory to practice

temic literacy to change their lifestyles or professional deliverables in a manner that demonstrates that.

When we make climate literacy the foundation of education, we are actually making an investment in social infrastructure. Having a climate-literate populace is a huge, decentralised carbon reduction engine. They make rational decisions regarding the use of energy, circular economies, and less waste because they know how the crisis works. But climate education in India has long been relegated to the Environment Studies (EVS) period. The government

should mainstream climate literacy in all fields to make a tangible change. Economics students ought to be computing carbon tax and social cost; Engineering students ought to be learning lifecycle analysis, and History students ought to be learning how resource management and the decline of civilisations relate to one another.

Corporate perspective

But the least considered stakeholder in this learning ecosystem is the corporate sector. With the growing pressure to achieve ESG targets and navigate through international carbon border adjustments,



GETTY IMAGES/ISTOCKPHOTO

Indian companies are finding a gigantic green skills gap. They require workers who not only understand that climate change is a reality but also how to decarbonise a supply chain or operate a sustainable investment portfolio. When the learning system is unable to do this, then the corporate world loses productivity. Hence, corporates need to stop consi-

dering climate education as a philanthropic gift and begin considering it as preparation for the market.

Alliances with universities and vocational institutes will allow businesses to make sure that future employees are prepared with the technical and cognitive instruments needed to achieve a low-carbon economy. This includes es-

tablishing living labs where students will be able to apply real-world sustainability issues encountered by industry. When a student spends his last year project working on a waste-management bottleneck at a local textile factory, he has left the theory behind to go into practice. They are no longer a bystander of the climate crisis but a participant in climate solutions.

Agents of change

Moreover, we should also look at how the youth can be part of the solution as present agents of change. This multiplier effect is precisely why education is a “high-leverage” mitigation strategy. It forms a bottom-up force that balances out top-down policy so that when government chooses an ambitious goal, such as Net Zero by 2070,

there is an educated and eager population to implement it.

The last challenge in this path is the unavailability of financing. We must transform our financial measures to understand that investment in climate education today can save money in disaster relief tomorrow. If we are subsidising a wind turbine, we can and must subsidise the training of the engineer who will maintain it and the policymaker who will regulate it. When all citizens have the power to act, the long-term pathways of emissions that we are currently terrified by will start to curve towards a more sustainable and strong future.

The real strength of climate education is how it will transform climate anxiety into climate action so that the leaders of tomorrow not only have heard about the fire but also have been trained to put it out.

The writer is an academician and researcher in the field of Economics and Business Policy at FORE School of Management.



GETTY IMAGES/ISTOCKPHOTO

Are you valued?

Why ‘mattering’ matters



THINK
Aruna Sankaranarayanan

Given that we live in a fractured world, it's not surprising that many people are living with a sense of unease. With mental health issues on the rise, individuals are contending with an existential crisis on an unprecedented scale. When young people feel hopeless rather than hopeful, it's a sign that something fundamental is amiss.

Jennifer Breheny Wallace discusses the concept of mattering in her recently published and timely book of the same name. In essence, mattering is about feeling that you are valued by others and that you can add value to the world. First coined by sociologist Morris Rosenberg, mattering is now “emerging as one of the most essential, yet overlooked” facets of well-being. Further, it is conceptualised as a “meta-need” as it factors in various psychological needs of “connection, belonging and purpose.”

Wallace avers that optimal well-being ensues only when this trio of needs are met. You may feel you belong to a family or a company but you don't feel you matter to those groups. Nobody cares about your opinion or asks how you're doing. While your work may be considered meaningful, you're unsure of the actual impact you're having. Caring professions like

medicine and teaching are typically perceived as being inherently fulfilling. Yet, many teachers and doctors feel their jobs are a grind as they don't often receive heartfelt appreciation from students or patients.

Similarly, though parenting is supposed to be inherently rewarding, when push comes to shove, stay-at-home parents may feel frustrated as neither their spouse nor their children seem to notice them.

Five elements

Wallace argues that people who are flourishing in life have a “mattering core” that serves as a buffer to navigate the twists and turns of life. Five elements constitute our mattering core. The first involves Recognition, where you feel that both “you and your actions” are valuable to others and that your “absence” would be missed by them. Small acts of appreciation can go a long way in contributing to a sense of recognition. When we thank a colleague for chipping in or a friend for keeping a coffee date during a busy spell, we contribute to building their mattering core.

The second facet is Reliance, where we “feel needed” as others are dependent on us. Whether it's caring for a child, an elderly relative or a pet, when others rely on us, it contributes to our mattering. If you feel you are just a cog in a wheel at work, you may consider volunteering or helping others on a regular basis. Ideally, try to employ your strengths to meet a need in the world.

While others may be

reliant on us, we also need to strike a “balance between adding value” and “feeling valued ourselves.” In fact, when others are over-reliant on us, it can lead to stress and burnout. The next ingredient of mattering is Importance, where you feel “prioritised” by other people. When your needs take precedence over others' by some people, at least some of the time, it makes you feel significant.

Ego Extension, the fourth element, refers to the sense of caring you experience when others share in your triumphs and travails. Usually, this feeling is reciprocal. When someone mirrors your feelings, it's very likely that you reflect theirs too.

The final ingredient, Attunement, means that you “feel deeply understood and meaningfully responded to.” It involves being supported in the moment in ways that meet our current needs. When a person can figure out, without asking, whether we need a hug or a word of encouragement or mere silence, they are attuned to us.

Thus, mattering is something we co-create with others. Just as others can foster or foil our sense of mattering, we too can bolster or belittle theirs. If families, schools, and workplaces cultivate a mattering culture that respects the inner lives of their people, both individuals and communities stand to gain.

The writer is visiting faculty at the School of Education, Azim Premji University, Bengaluru, and the co-author of Bee-Witched.

Shrinwanti Shee

The Common University Entrance Test (CUET-UG), conducted by the National Testing Agency (NTA), is a national exam for admission into undergraduate and diploma programmes across central and participating universities in India and plays a crucial role in securing admission to top institutions.

Subject selection

One of the most important aspects of CUET is subject selection. Applicants can select up to five subjects from a pool of 37 across three sections. Students focus on selecting individual subjects but universities usually evaluate subject combinations that reflect academic direction, determine eligibility for specific programmes, and open (or limit) future opportunities, especially in interdisciplinary fields. So here's how to choose subjects for the CUET.

A structured approach can significantly improve



GETTY IMAGES/ISTOCKPHOTO

Building blocks

Tips to choose the right subjects for CUET

your chances of success. First, ensure that the subjects you choose are aligned to the programme you wish to pursue. Ensure you meet the requirements and eligibility criteria for the institution you are targeting. Select the language you will answer in wisely. English is widely required. Understand the paper structure and sections and try to opt for cross-disci-

plinary subjects to keep multiple career options open. Certain combinations offer strong academic and career advantages.

For example, Maths, Economics, and Accountancy are ideal for careers in Analytics and Business. A Physics, Chemistry, and Maths combination will open pathways in Engineering, Data Science and in the non-Engineering do-



GETTY IMAGES/ISTOCKPHOTO

Personalised prep

How technology helps effective learning in NEET preparation

ration” look like? He/she would hire the best tutor available who would study the student carefully, identify strengths and weaknesses, design customised questions that target specific gaps, maintain detailed progress reports and conduct frequent tests. In short, the tutor would deliver a personalised, data-

driven, continuous learning intervention. The catch? This model was expensive, scarce, and inaccessible to most. Technology has now democratised this ecosystem. Today, personalisation is no longer a luxury.

Today, a NEET aspirant can choose from top educators in the online space

and learn at their own pace or revisit lectures, pause, rewind, and, unlike real life, never hesitate to “ask again”. But, more importantly, technology now builds a learning system around the student. This means that a student can generate summaries of any lesson at a desired depth and generate customised tests with precise control over subjects, difficulty levels, and question types or practice sets to target weak spots.

Revealing insights

Every mistake is not just corrected; it is captured, classified, and remembered. An automated error log maintains a personalised record of weaknesses. Over time, the system begins to “understand” the learner: which topics consume more time; which question types cause hesitation, and where accuracy drops under pressure. Students can compare their performance with peers, analyse time efficiency, and identify patterns that would otherwise remain

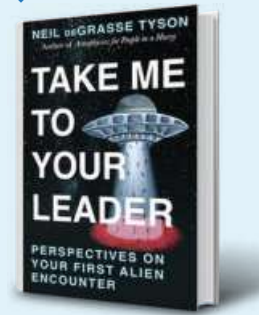
invisible. Next comes the reinforcement layer, which involves using flashcards for active recall, podcasts for passive learning, smart reminders, and gamified elements to sustain engagement.

Since no two students prepare the same way, each learner's dashboard, recommendations, tests, reminders, and revision plans are uniquely shaped by their behaviour, performance, and gaps. The system adapts continuously, quietly correcting, nudging, and optimising.

Technology, therefore, does not replace effort, discipline, or conceptual clarity. Instead, it ensures that time is spent where it is needed most, weaknesses are not ignored but systematically addressed, and progress is measurable, visible, and improvable. The fundamentals of learning remain unchanged, but the path to mastering them can be uniquely yours.

The writer is National Academic Director-Medical, Aakash Educational Services Limited.

ON THE SHELF



■ **Take Me To Your Leader**
Would an alien really be a humanoid with a pointy head? How should you communicate with an alien who doesn't speak English? Why has no alien (apparently) ever contacted us? If an alien did demand 'Take me to your leader!', who are you actually meant to take it to? This book is both a work of popular science and a love letter to our as-yet-uncontacted

extraterrestrial friends by one of the best-loved scientists in the world and, perhaps, in the galaxy.
Author: Neil deGrasse Tyson
Publisher: Penguin
Price: ₹999

■ **We the People**
The US Constitution is 237 years old. It has been amended just 27 times. Nearly 12,000 amendments have been proposed; almost all

have failed. This book, winner of the 2026 Pulitzer Prize for History, follows the ordinary citizens and people who fought to make the document live up to what it promised. The point is: the Constitution was never supposed to be permanent. It was expected to evolve. That it hasn't is not a sign of strength but a warning.
Author: Jill Lepore
Publisher: Hachette
Price: ₹1599

